

- 4 Fig. 4.1 shows two identical solenoids with their axes aligned and the two opposite faces at a distance of 0.20 m apart.

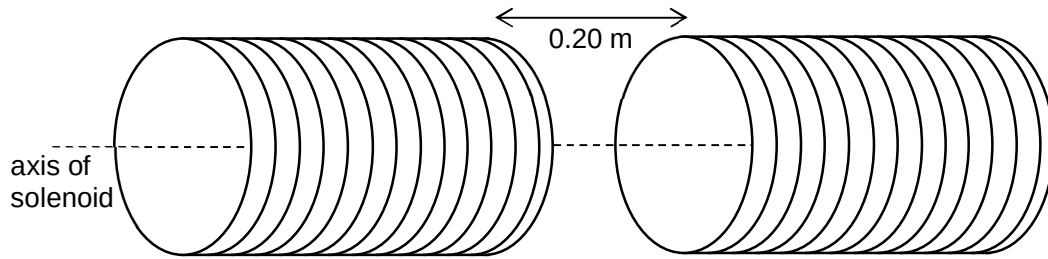


Fig. 4.1

The space between the solenoids is a vacuum with a uniform magnetic flux density of $1.2 \times 10^{-3} \text{ T}$.

An electron moving with speed $5.0 \times 10^7 \text{ m s}^{-1}$ normal to the magnetic field between the solenoids traces an arc of a circular path.

- (a) (i)** Determine the radius of the circular path.

radius = m [2]

- (ii)** State and explain whether the electron will collide with any of the solenoids.

.....

 [2]

- (b)** In another similar experiment described above, the electron is found to spiral inwards with decreasing radius even though the magnetic field remains constant.

With reference to the speed of the electron, suggest a reason for this motion.

.....
.....
..... [2]

- (c)** A uniform electric field is applied in the region of the magnetic field so that the electron continues undeflected through the two fields.

Determine the magnitude of the electric field strength E .

$$E = \dots\dots\dots \text{ N C}^{-1} \text{ [2]}$$